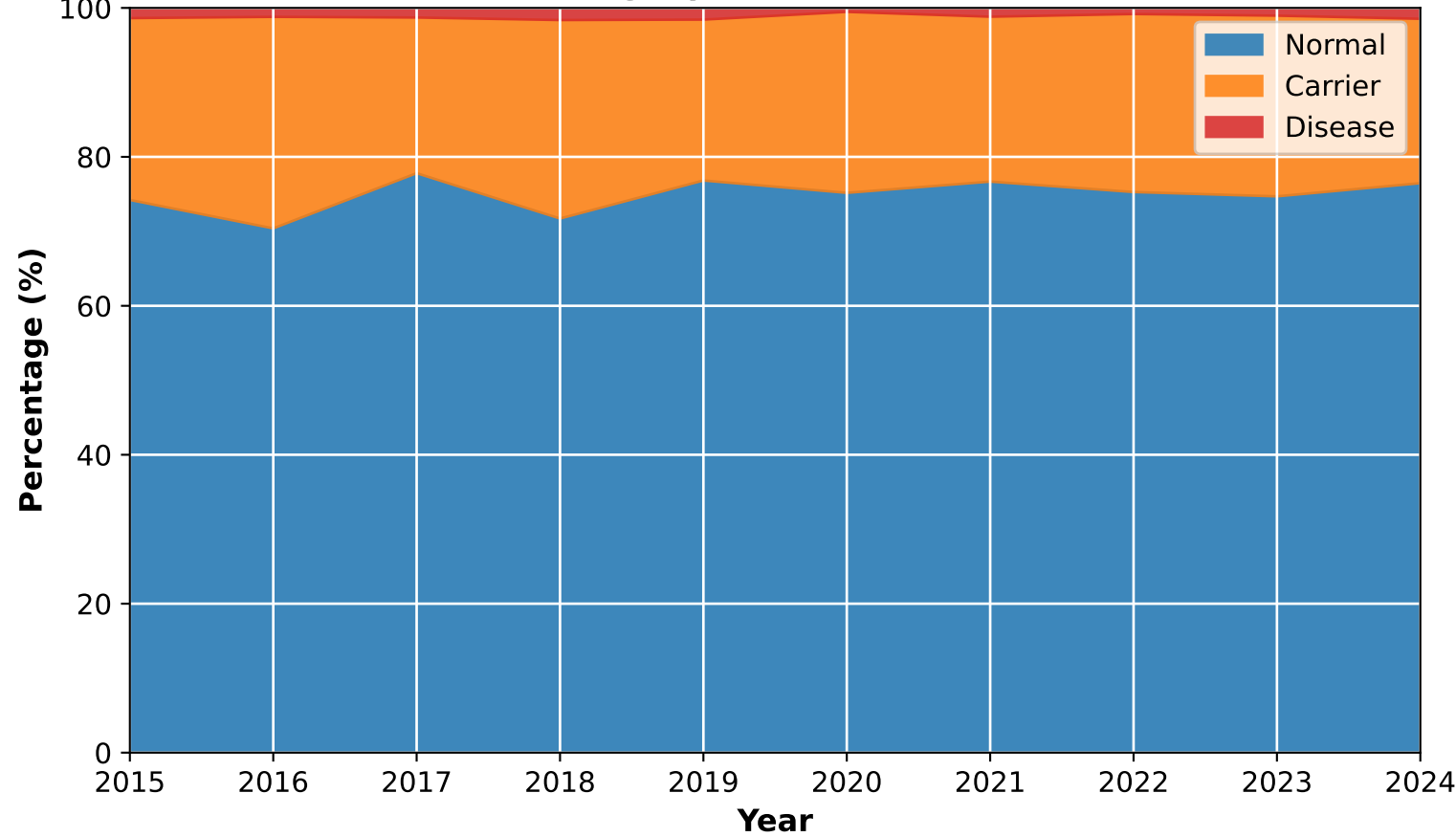
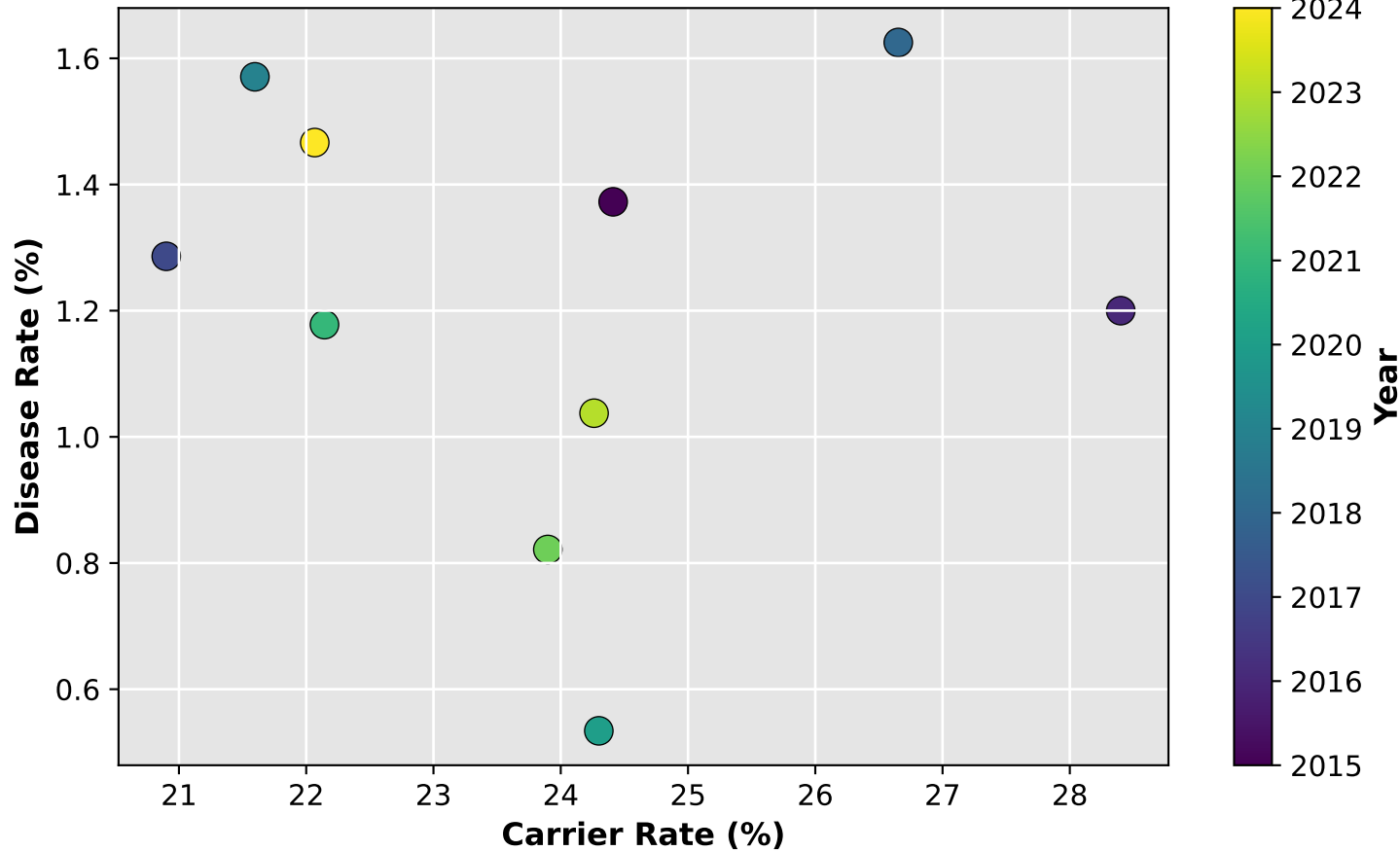


# Supplementary Analysis: Comprehensive Clustering Analysis Results

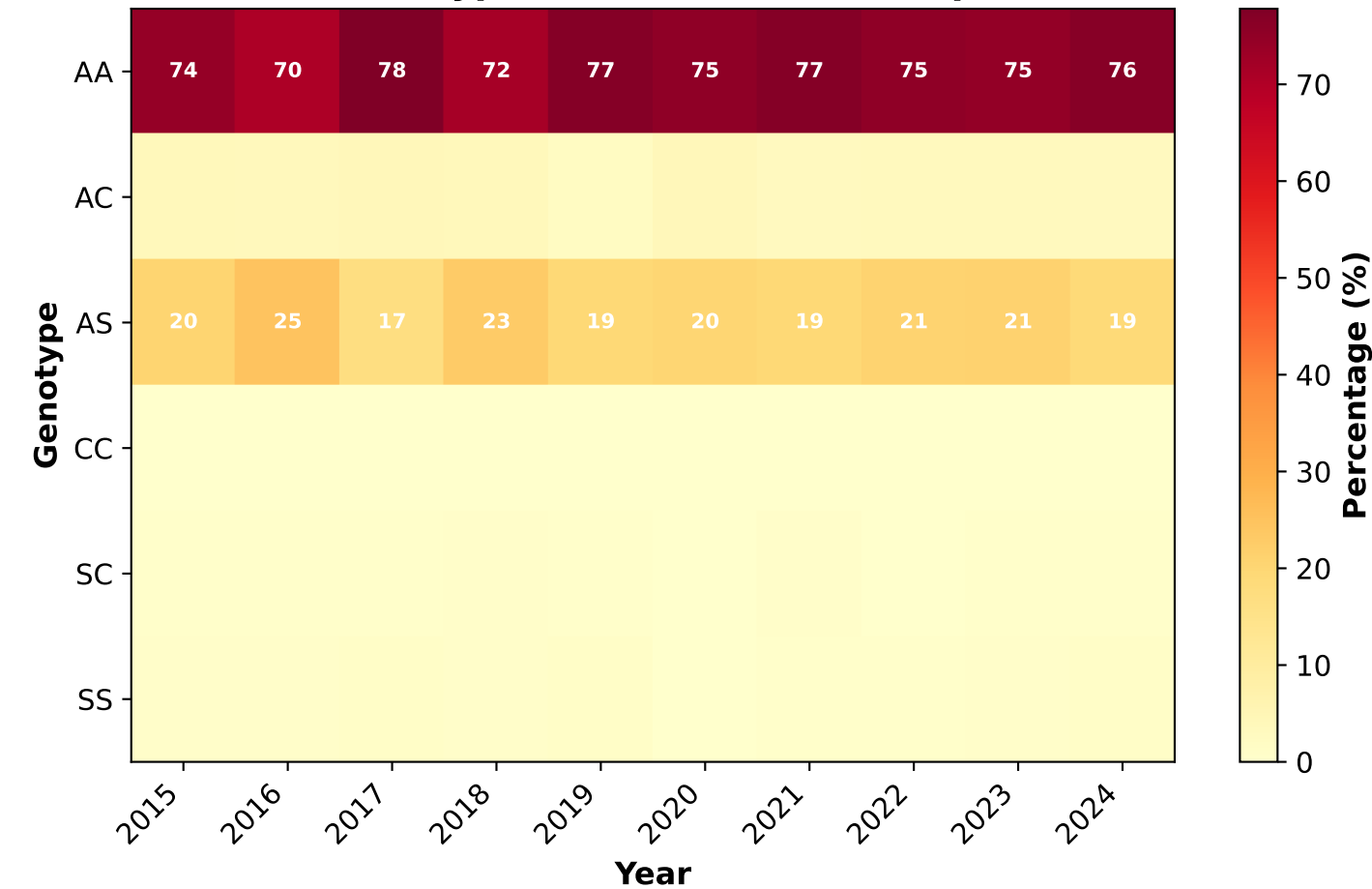
Clinical Category Distribution Over Time



Carrier vs Disease Rate Relationship



Genotype Distribution Heatmap



## Clustering Analysis Summary:

Total Samples: 8,890  
Years Analyzed: 10

### Genotype Distribution:

- AA (Normal): 75.1%
- AS (Carrier): 20.4%
- AC (Carrier): 3.3%
- SS (Disease): 0.7%
- SC (Disease): 0.5%
- CC (Disease): 0.0%

### Clinical Categories:

- Normal: 75.1%
- Carrier: 23.7%
- Disease: 1.2%

### Key Finding:

K-means clustering reveals three distinct temporal patterns, validating the heterogeneity within the seemingly stable population structure.